

A benchmark study of existing algorithms for mapping long reads

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Supplementary Tables

Table S1. Running command lines for each mapping method.

Methods	Running command lines
BWA-MEM	bwa index ref.fa bwa mem -t 48 -x pacbio ref.fa reads.fa > map.sam bwa mem -t 48 -x ont2d ref.fa reads.fa > map.sam
Graphmap	graphmap align -t 48 -r ref.fa -d reads.fa -o map.sam
Kart	kart index ref.fa ref kart -t 48 -pacbio -i ref -f reads.fa -o map.sam
NGMLR	ngmlr -t 48 -r ref.fa -q reads.fa -o map.sam
minimap2	minimap2 -t 48 -a -x map-pb ref.fa reads.fa > map.sam
lordFAST	lordfast --index ref.fa lordfast --threads 48 --search ref.fa --seq reads.fa > map.sam
mapAlign	mapAlign align 48 15 ref.fa reads.fa map.sam
lra	lra index genome.fa lra align ref.fa reads.fa -t 48 -p s > map.sam
os-minimap2	minimap2 -t 48 -a --syncs 11 --synct 3 ref.fa reads.fa > map.sam
BWA-MEME	bwa-meme index ref.fa bwa-meme mem -t 48 ref.fa reads.fa -o map.sam
Winnowmap2	meryl count k=15 output merylDB ref.fa meryl print greater-than distinct=0.9998 merylDB > repetitive_k15.txt winnowmap -t 48 -W repetitive_k15.txt -a -x map-ont ref.fa reads.fa > map.sam
mm2-fast	minimap2 -t 48 -a -x map-pb ref.fa reads.fa > map.sam
PSS	minimap2 -t 48 -a -x map-pb -k 15 -w 50 --downsample 1.23 --s-mer 5 --pos1 1 --pos2 10 ref.fa reads.fa > map.sam
VACmap	conda activate vacmap vacmap -ref ref.fa -read reads.fa -t 48 -mode H > map.sam
invMap	invmap ref.fa reads.fa > map.sam
pathMap	pathmap ref.fa reads.fa > map.sam
MashMap	mashmap -t 48 -s 500 -r ref.fa -q reads.fa > map.txt
MUMmer4	nucmer -t 48 ref.fa reads.fa -p out show-coords out.delta > out.coords

Table S2. Running command lines and parameter settings of SURVIVOR and PBSIM2 software for generating simulated datasets with SVs.

Simulators	Command lines
SURVIVOR	SURVIVOR simSV parameter_file SURVIVOR simSV chr1.fa parameter_file 0 1 chr1_with_sv
PBSIM	pbsim --data-type CLR --length-min 1000 --depth 40 --prefix simulated chr1_with_sv.fa

Table S3. The number of correctly mapped reads that cover inversion, the total number of extracted reads with inversion is 5864.

Methods	Total mapped	Correctly covered	Sensitivity (%)
BWA-MEM	5862	3047	51.96
GraphMap	5778	170	2.89
Kart	5864	276	4.70
NGMLR	5851	5514	94.03
minimap2	5863	5262	89.73
lordFAST	5864	2020	34.44
mapAlign	5842	323	5.50
lra	5855	4803	81.90
os-minimap2	5859	5335	90.97
BWA-MEME	5312	52	0.88
Winnowmap2	5762	5104	87.03
mm2-fast	5863	5262	89.73
PSS	5864	5242	89.39
VACmap	5857	5612	95.70
invMap	5864	5619	95.82
pathMap	5864	5574	95.05

Table S4. The number of correctly mapped reads that cover inversion, the total number of extracted reads with translocation is 11471.

Methods	Total mapped	Correctly covered	Sensitivity (%)
BWA-MEM	11470	6205	54.09
GraphMap	11235	366	3.19
Kart	11471	569	4.96
NGMLR	11388	9876	86.09
minimap2	11470	1039	9.05
lordFAST	11471	3848	33.54
mapAlign	11428	633	5.51
lra	11453	580	5.05
os-minimap2	11469	1271	11.08
BWA-MEME	10447	98	0.85
Winnowmap2	11306	1901	16.57
mm2-fast	11470	1039	9.05
PSS	11471	917	7.99
VACmap	11463	8881	77.42
invMap	11471	574	5.00
pathMap	11471	1016	8.85

Table S5. The number of correctly mapped reads that cover inversion, the total number of extracted reads with duplication is 5933.

Methods	Total mapped	Correctly covered	Sensitivity (%)
BWA-MEM	5742	2	0.03
GraphMap	5733	0	0
Kart	5742	1	0.01
NGMLR	5734	105	1.76
minimap2	5742	23	0.38
lordFAST	5742	1518	25.58
mapAlign	5731	2	0.03
lra	5735	94	1.58
os-minimap2	5740	20	0.33
BWA-MEME	5233	3	0.05
Winnowmap2	5664	41	0.69
mm2-fast	5742	23	0.38
PSS	5742	17	0.28
VACmap	5736	0	0
invMap	5742	0	0
pathMap	5742	20	0.33

Table S6. Running command lines and parameter settings for generating two long sequences with different error rates. This simulator is designed in WAF2 paper.

Simulator	Command lines
generate_dataset	generate_dataset --n 1 --length 1000000 --error 0.01 ref.fa --o out.txt

Table S7. Run time (s) of pairwise sequence alignment for different sequence lengths with 99% similarity.

Similarity (%)	Methods	Sequence lengths				
		10^6	5×10^6	10^7	5×10^7	10^8
99	minimap2	0.42	2.45	3.85	24.57	53
	os-minimap2	0.35	1.68	3.38	16.9	47.12
	Winnowmap2	2.38	4.9	7.3	25.63	51.62
	mm2-fast	0.41	2.52	4	25.49	53.34
	PSS	0.55	2.99	7.73	219	1456
	invMap	1.03	5.49	11.2	75.1	208
	pathMap	0.93	9.86	20	114.44	273
	Edlib	3.24	54.04	203.72	4197	16697
	libgaba	0.12	0.64	1.28	6.2	12.37
	WFA2	0.38	1.99	3.93	21	38.55
90	minimap2	0.42	2.05	4.06	20.25	52
	os-minimap2	0.4	1.87	3.71	18.82	47.78
	Winnowmap2	3.18	6.04	9.32	38.96	52.15
	mm2-fast	0.43	2.06	4.1	20.5	52.26
	PSS	0.48	2.62	6.73	196	1451
	invMap	0.75	4.1	8.13	55	214
	pathMap	0.68	4.9	10.31	61.5	273
	Edlib	20	413	1637.44	16682	57875
	libgaba	0.16	0.68	1.4	6.82	12.65
	WFA2	3.25	19.15	19.15	40.32	64.64
80	minimap2	0.72	3.63	7.2	35.14	51.64
	os-minimap2	0.61	2.93	5.87	29.17	46.71
	Winnowmap2	4.83	14.44	26.6	52.37	124.05
	mm2-fast	0.72	3.56	7.44	35.99	52.82
	PSS	0.59	3.1	7.41	193	1454
	invMap	0.71	3.87	7.79	47.53	207
	pathMap	0.69	3.92	7.92	48.81	272
	Edlib	38.69	808.37	3222.85	16653	114875
	libgaba	0.19	0.77	1.49	7.25	12.52
	WFA2	5.74	30.63	30.63	43.78	74.13

Table S7. Run time (s) of pairwise sequence alignment for different sequence lengths with 99% similarity (Continued).

70	minimap2	2.42	11.54	22.8	52.09	109.48
	os-minimap2	0.89	5.57	10.69	48.37	52.43
	Winnowmap2	5.96	21.91	39.1	52.33	188.12
	mm2-fast	2.41	10.80	21.13	54.31	102.43
	PSS	1.05	5.48	11.74	212	1455
	invMap	0.95	4.96	10.3	59.07	209
	pathMap	0.92	4.88	10.04	57.35	285
	Edlib	43.25	1481.46	5908	16655	124234
	libgaba	0.2	0.77	1.52	7.56	12.43
	WFA2	7.13	42.83	42.83	49.09	161.54
60	minimap2	1.78	9.66	20.97	52	95.22
	os-minimap2	1.63	8.58	16.44	47.22	83.33
	Winnowmap2	9.1	34.85	52.12	66.93	316.41
	mm2-fast	1.65	8.42	18.77	52.83	82.62
	PSS	2.78	12.36	24.38	275	1460
	invMap	1.67	8.64	17.74	96.79	366
	pathMap	1.76	8.57	17.29	93.55	286
	Edlib	75.10	1574.86	6294	16653	133990
	libgaba	0.21	0.81	1.59	7.82	12.71
	WFA2	8.68	39.4	94.82	94.82	189.09

Table S8. Memory usage (MB) of pairwise sequence alignment for different sequence lengths.

Similarity (%)	Methods	Sequence lengths				
		10^6	5×10^6	10^7	5×10^7	10^8
99	minimap2	22.75	98.84	194.34	1162.28	2093.53
	os-minimap2	14.33	60.29	137.48	735.64	2396.33
	Winnowmap2	1049.50	1055.57	1079.98	1181.84	1424.41
	mm2-fast	23.31	102.21	215.71	1076.96	1914.75
	PSS	23.05	102.79	194.66	1434.89	4350.46
	invMap	138.17	598.25	1330.12	4352.29	6280.12
	pathMap	139.85	665.17	1334.26	6279.14	12265.71
	Edlib	11.50	47.97	94.25	466.08	823.80
	libgaba	83.25	408.37	814.50	4066.87	8132.06
	WFA2	399.37	1999.68	1999.68	3990.93	40191.76
90	minimap2	18.26	80.2890625	193.4453125	932.6015625	2096.351563
	os-minimap2	15.45	63.0859375	127.4453125	687.6796875	2399.824219
	Winnowmap2	1049.42	1060.109375	1070.617188	1161.789063	1430.296875
	mm2-fast	20.515625	87.85546875	198.3085938	931.53125	1913.628906
	PSS	20.796875	88.1875	186.96875	1290.21875	4348.8125
	invMap	94.328125	493.2734375	1016.039063	4673.414063	12166.68359
	pathMap	94.31640625	543.921875	1080.65625	4767.589844	12165.81641
	Edlib	12.09765625	46.0234375	93.328125	445.8164063	824.5273438
	libgaba	83.25	408.375	814.5	4066.875	8131.5
	WFA2	3362.0625	20618.74219	20618.74219	39865.37109	61109.23438
80	minimap2	43.0078125	101.65625	266.9570313	956.265625	2095.28125
	os-minimap2	19.38671875	72.19140625	144.8398438	690.65625	2406.550781
	Winnowmap2	1049.683594	1059.351563	1072.195313	1173.152344	1405.089844
	mm2-fast	45.8203125	104.3085938	263.5742188	994.125	1917.0625
	PSS	32.6328125	108.4492188	187.984375	1270.074219	4349.253906
	invMap	85.96875	449.2070313	911.1328125	4003.628906	12301.51953
	pathMap	86.53515625	451.4179688	892.1132813	4144.355469	12302.32422
	Edlib	11.5625	46.890625	92.046875	466.6289063	824.7265625
	libgaba	83.25	408.375	815.0625	4066.875	8131.5
	WFA2	5794.3125	32617.15625	32617.15625	43272.14453	81209.50391

Table S8. Memory usage (MB) of pairwise sequence alignment for different sequence lengths (Continued).

70	minimap2	416.515625	502.6796875	589.9453125	1495.070313	2097.121094
	os-minimap2	33.56640625	138.4726563	255.0898438	764.8867188	2399.195313
	Winnowmap2	1049.515625	1301.792969	2490.53125	1411.171875	2956.277344
	mm2-fast	419.328125	505.4804688	593.8828125	1664.964844	1918.890625
	PSS	83.24609375	187.1679688	324.1367188	2660.894531	4352.292969
	invMap	97.76953125	487.8476563	956.6015625	4041.601563	12265.73828
	pathMap	101.1445313	478.6015625	945.390625	4132.96875	12264.38672
	Edlib	11.7109375	44.28515625	92.59375	419.9375	825.1054688
	libgaba	83.25	408.375	815.0625	4066.875	8132.0625
	WFA2	7599.9375	40346.61719	40346.61719	40438.90234	126714.793
60	minimap2	315.2304688	625.9570313	1076.328125	1775.613281	2097.433594
	os-minimap2	31.19140625	235.7890625	258.4023438	808.5117188	2409.382813
	Winnowmap2	1049.785156	1417.363281	1723.5625	2660.894531	5092.355469
	mm2-fast	318.6054688	630.2070313	1084.214844	1604.222656	1912.578125
	PSS	445.5	507.3007813	610.8632813	1282.117188	4349.480469
	invMap	136.40625	499.3867188	1003.871094	4055.640625	12175.27734
	pathMap	189.2851563	528.4140625	1009.828125	4151.625	12176.94141
	Edlib	12.16796875	43.4375	94.63671875	479.1640625	830.109375
	libgaba	83.25	408.375	814.5	4066.875	8132.0625
	WFA2	9012.207031	40261.22266	86591.85938	86591.85938	182636.4375

Supplementary Figures

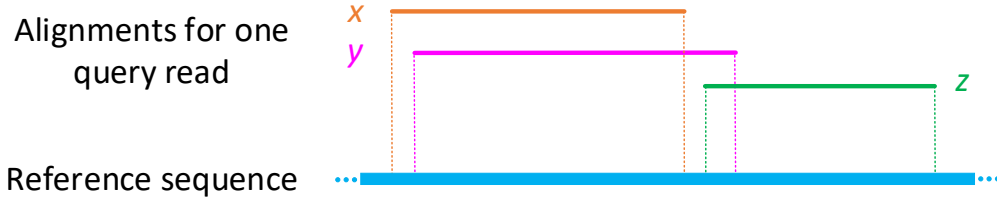


Figure S1. Examples of covering and non-covering alignment results. x , y and z are different alignment results reported by three mapping methods for the same query read. The dotted line indicates the mapped region in the reference genome sequence for the corresponding alignment result. In this figure, x and y alignments cover each other as they span the mapped regions on the reference genome that have at least 90% overlap. On the other hand, the alignment z does not cover either alignment x or y .

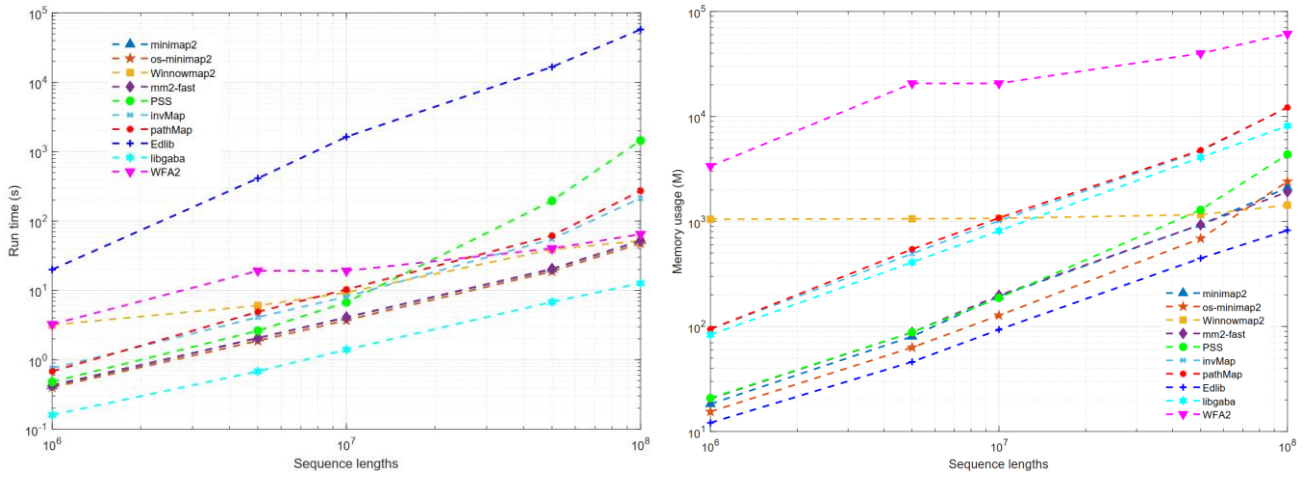


Figure S2. Run time and memory usage for pairwise sequence alignment for different sequence lengths with 90% similarity.

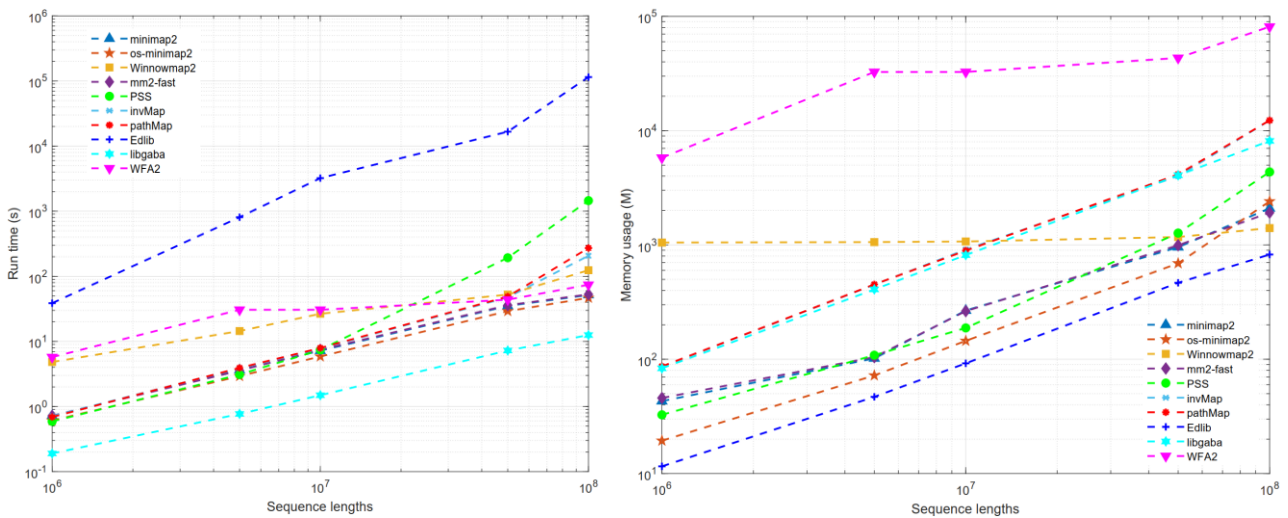


Figure S3. Run time and memory usage for pairwise sequence alignment for different sequence lengths with 80% similarity.

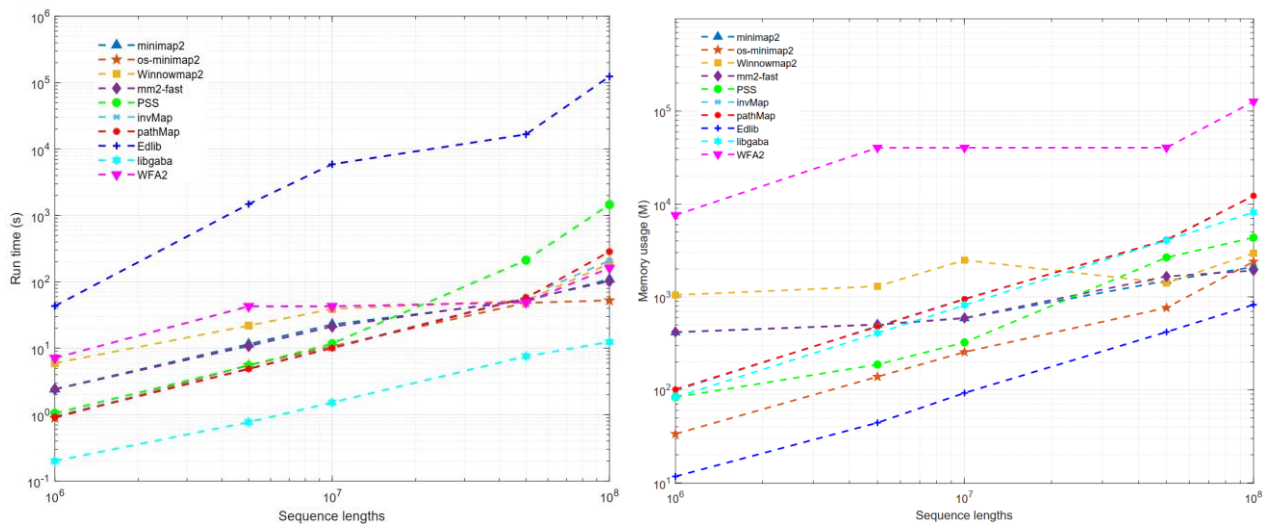


Figure S4. Run time and memory usage for pairwise sequence alignment for different sequence lengths with 70% similarity.

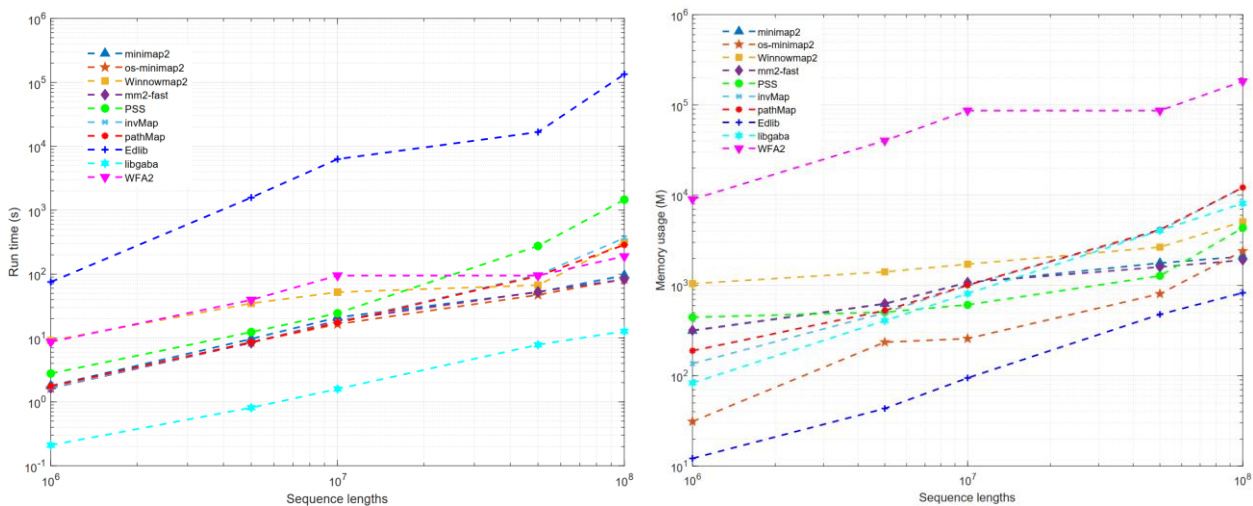


Figure S5. Run time and memory usage for pairwise sequence alignment for different sequence lengths with 60% similarity.